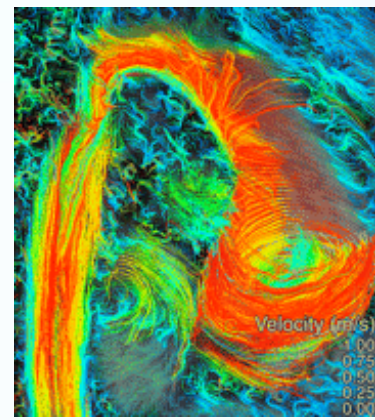
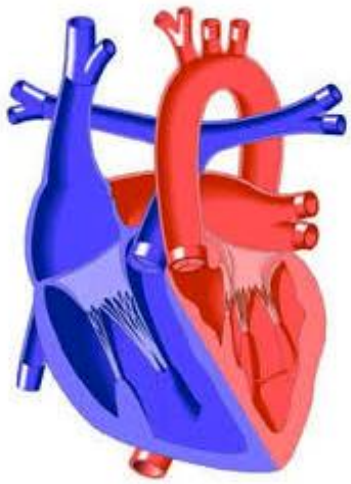




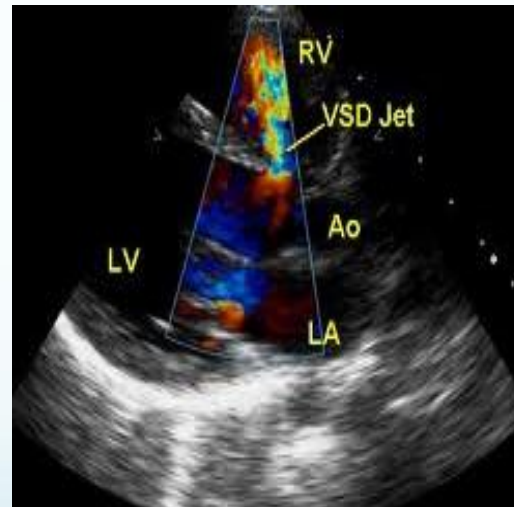
Adults with Congenital Heart Disease An Emerging Population

Dr. Rachael James MB BS, MD, BSc, FRCP
Consultant Cardiologist
Sussex Cardiac Centre
Brighton

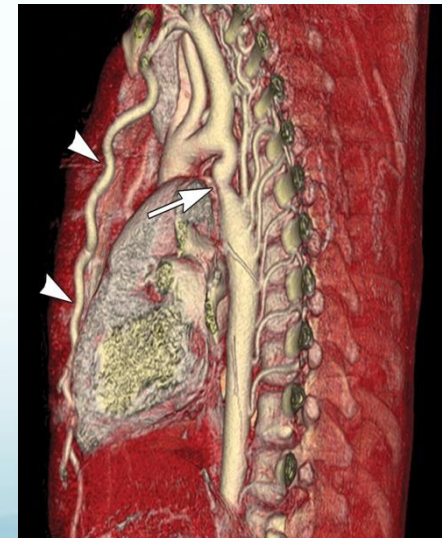
Simple Congenital Heart Defects

- Atrial septal defect
- Ventricular septal defect
- Aortic coarctation
- Pulmonary stenosis

VSD



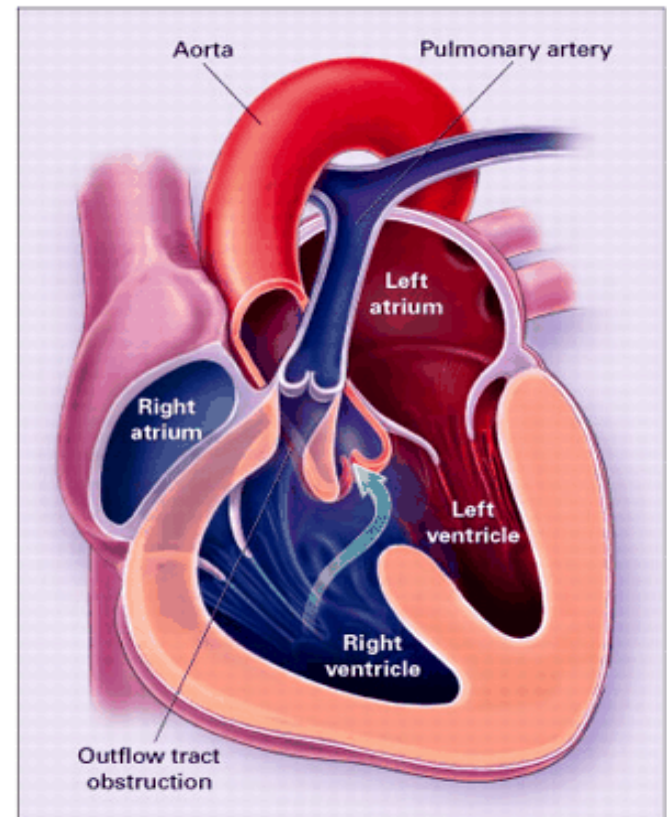
Aortic Coarctation



Complex Congenital Heart Defects

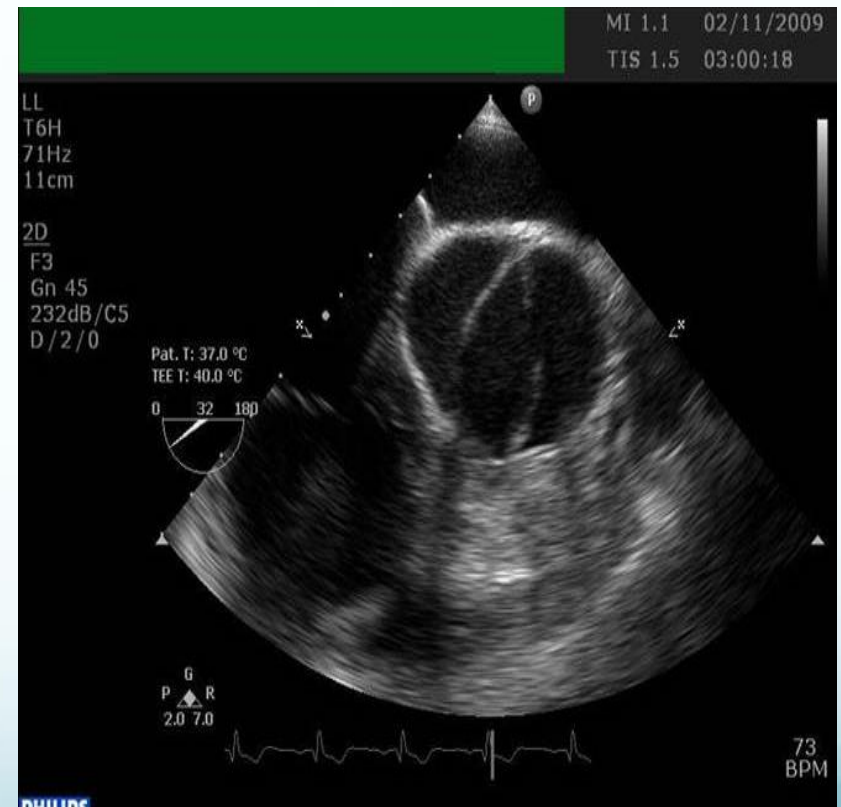
- Tetralogy of Fallot
- Atrio-ventricular canal defects
- Transposition of the great arteries
- Hypoplastic left ventricle

Tetralogy of Fallot



Incidence Congenital Heart Disease UK

- 8 per 1000 live births
 - 19 per 1000 if included bicuspid aortic valve
- 75% defects diagnosed by age 1
- 18% between ages 1-4 years
- cardiovascular malformations account >40% all deaths from congenital anomalies



Echo (TOE) image bicuspid aortic valve

Bicuspid Aortic Valve (BAV)

Wembley Stadium
capacity 90,000

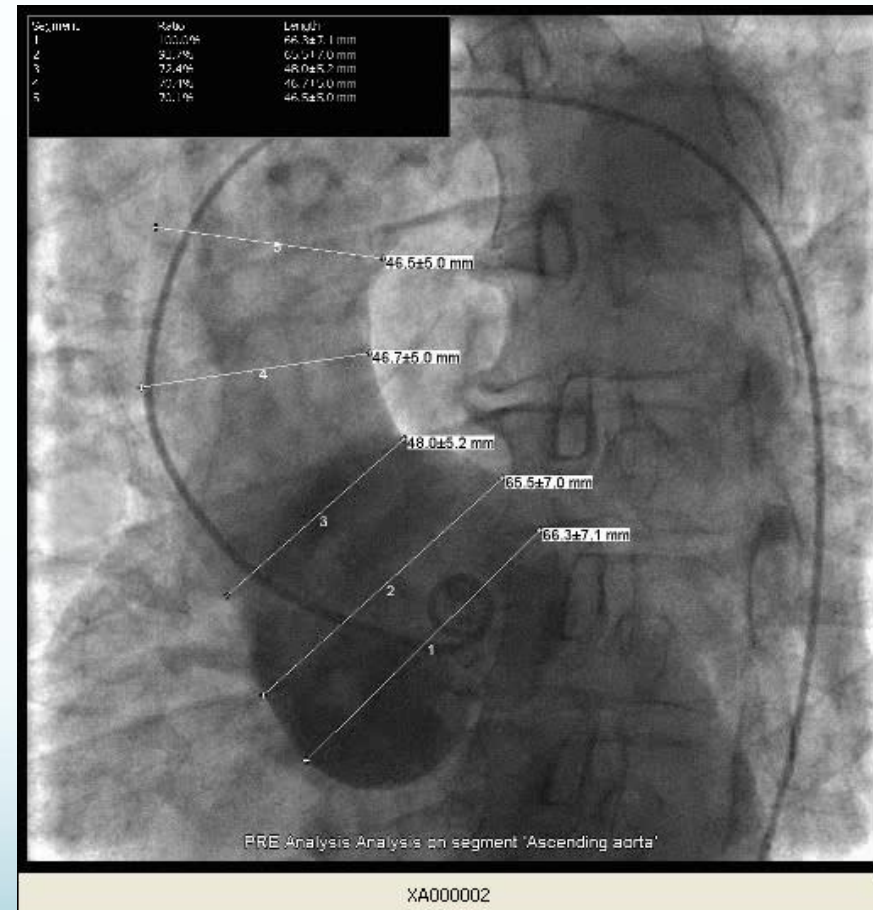
- Affects 1% population
- M:F 3:1
- Association Turner syndrome 45 XO
- 9% risk if 1st degree relative affected
- Commonest cause AVR in under 70s in Europe and North America
- 1:50 children clinically significant valve disease in adolescence



900 individuals in a full stadium will have a bicuspid aortic valve

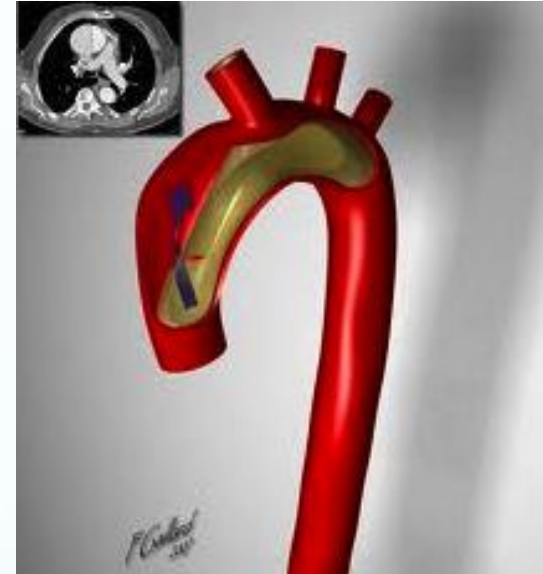
Bicuspid Aortic Valve

- understanding BAV still incomplete, characterised by:
 - abnormal valvulogenesis
 - aortic pathology
- 50% adults BAV have aortopathy
- Olmstead County study:
 - * 27% adults BAV & no significant valve disease @ baseline required surgery within 20 years follow up
 - * only 3% for aortic regurgitation
 - * 15% had co-existent ascending aortic dilation ($> 4\text{cm}$)
 - * root size related:
 - age, aortic regurgitation, male gender, systolic hypertension



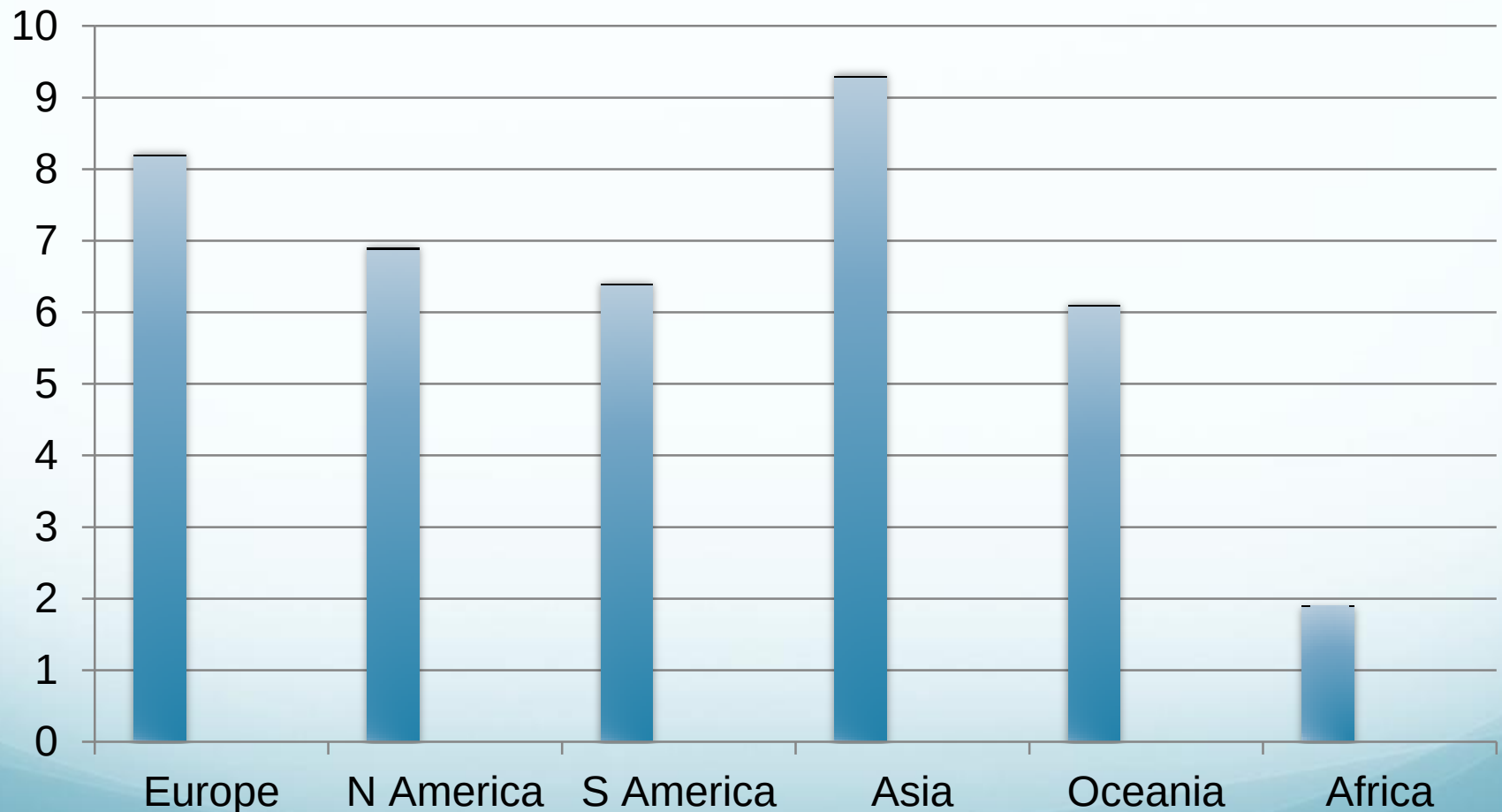
Bicuspid Aortic Valve & Dissection

- Presence BAV increases dissection risk 9 fold
- Risk lower than Marfan syndrome
 - BAV carries 6.1% lifetime risk compared 40% Marfan syndrome
- BAV 100 x more common than Marfan syndrome
- Incidence dissection BAV is equal or more common than Marfan syndrome

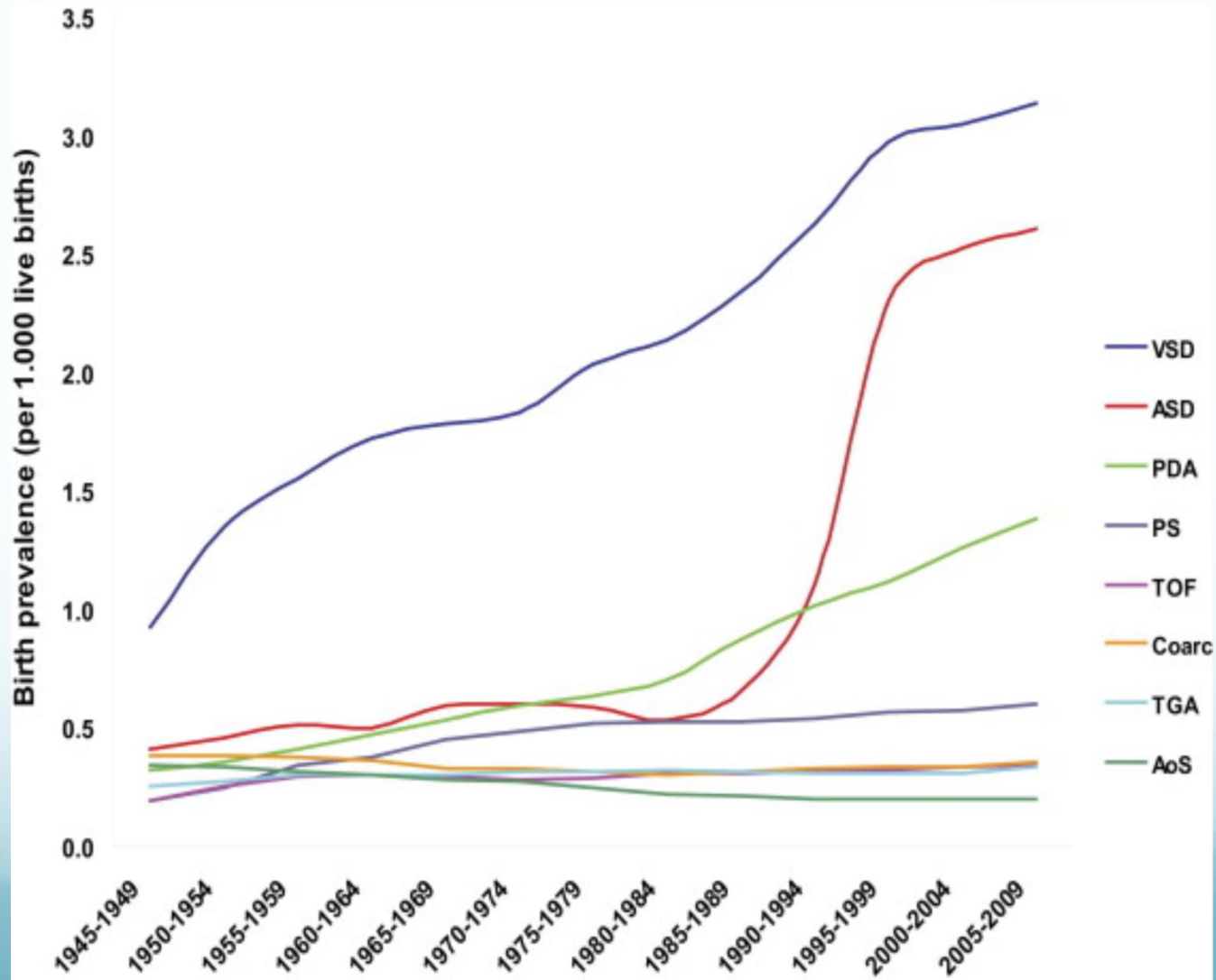


Risk factors dissection:
male gender
absolute aortic size (mean 5.9cm)
aortic regurgitation
L:R configuration

Global Birth Prevalence (per 1000 live births)



Global Birth Prevalence CHD Subtypes Over Time



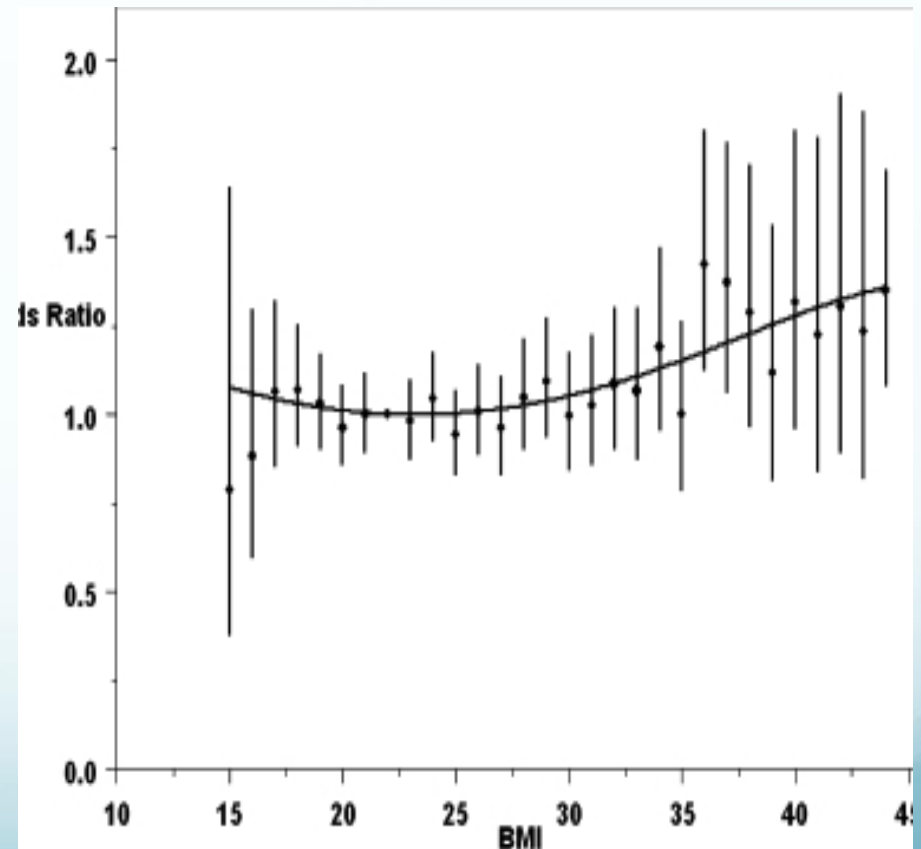
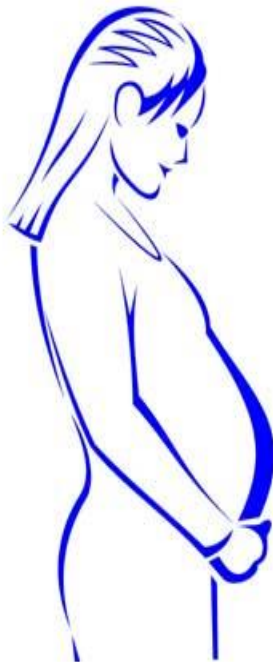
Why the Change?

- Increased premature baby survival
- Advancing maternal age
 - mean maternal age UK is 30
 - 20% mothers aged 35 or over (17% increase from previous decade)
- Maternal diabetes
 - incidence CHD in babies mothers IDDM 5 x general population
- Assisted reproduction (IVF)
- Urbanization / industrialization
- Marijuana use
- Exposure organic solvent
- Maternal obesity

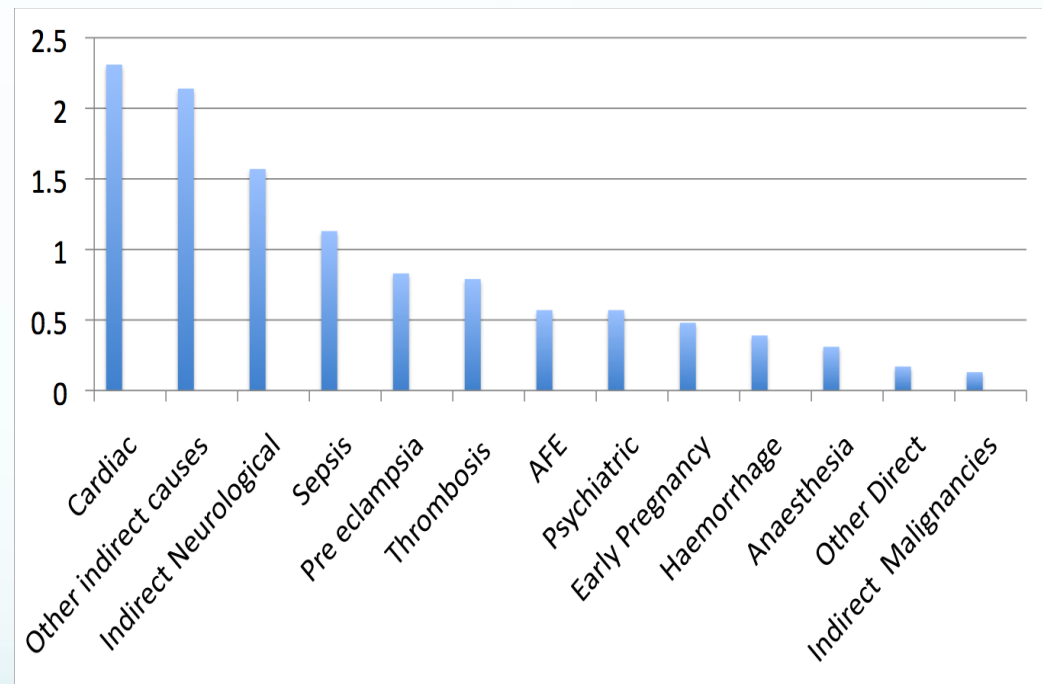
Maternal Obesity

Odds Ratios CHD by Maternal BMI

- 5% pregnant women in UK BMI >35



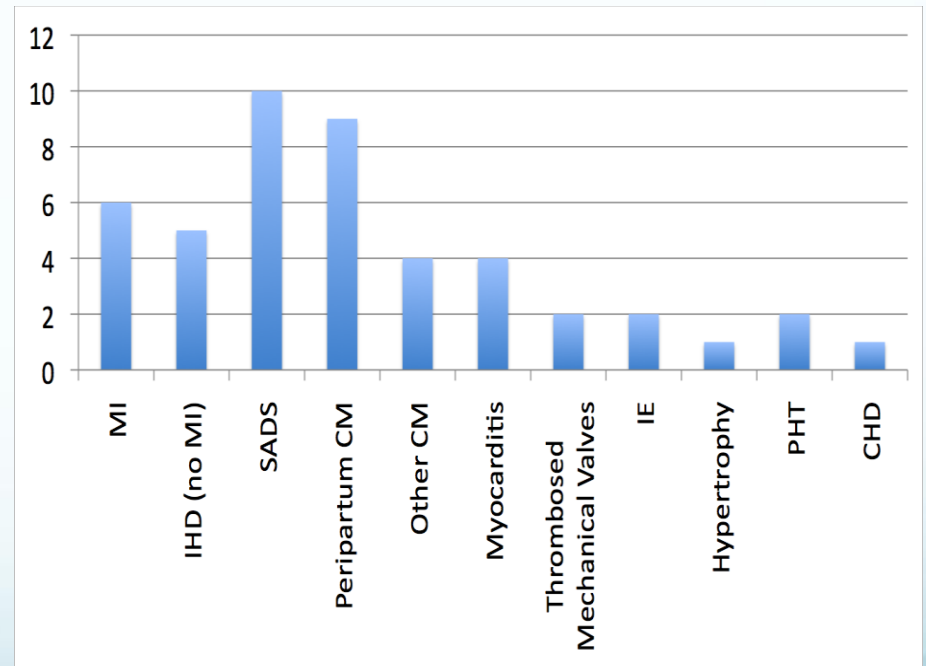
Leading Causes Maternal Death in UK 2006-08 Confidential Enquiry



Rates per 100 000 maternities

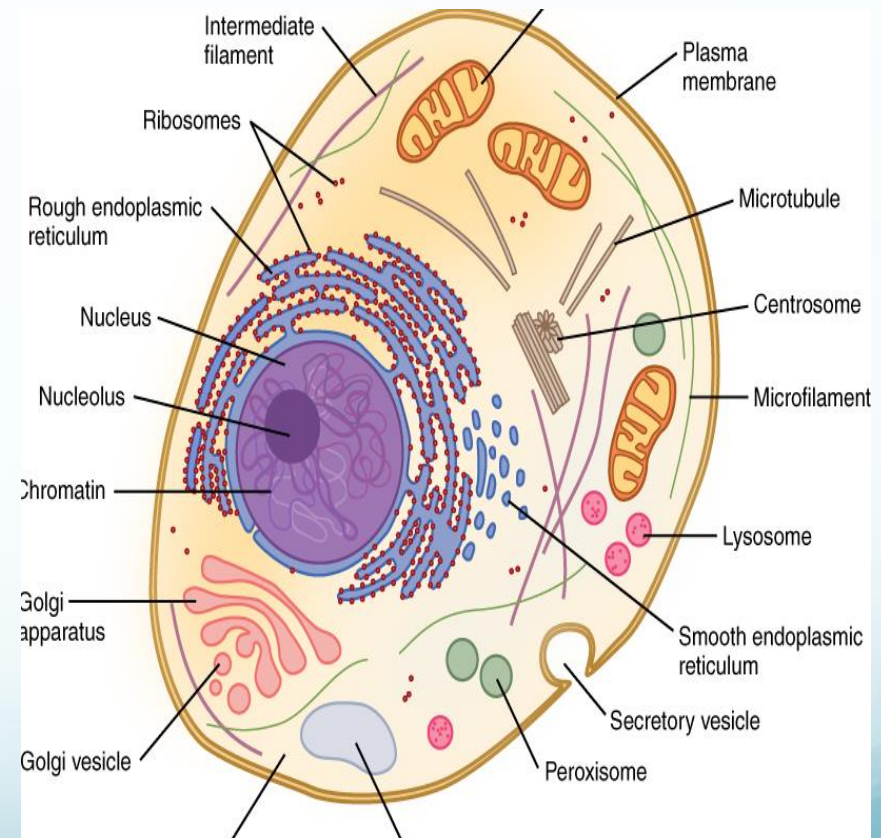
Cardiac Causes Maternal Death UK 2006-08

- 60% cardiac-related maternal deaths were in overweight or obese women
- Ischaemic heart disease commonest cause death



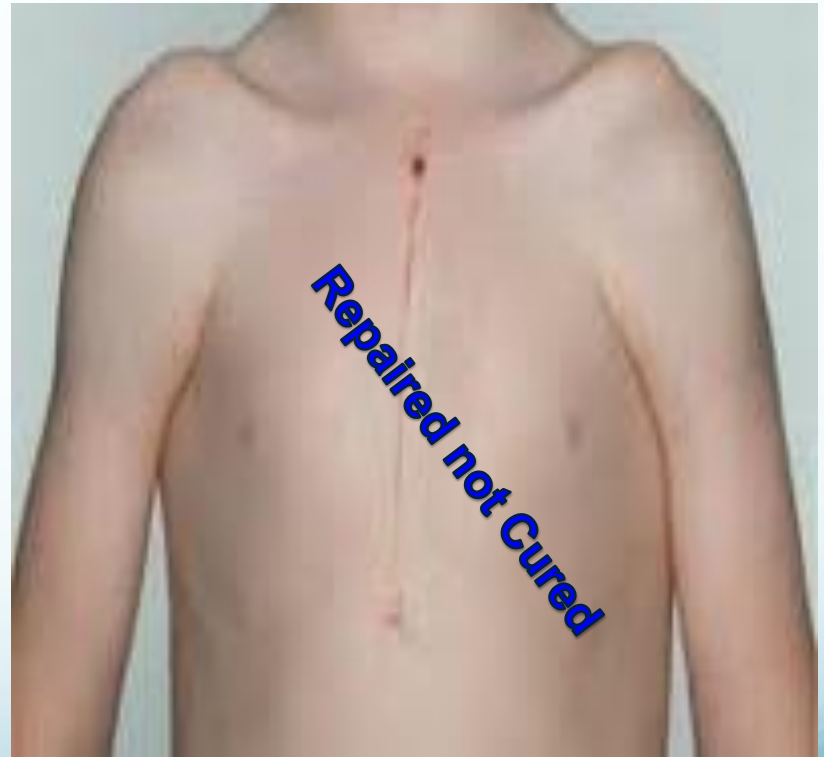
Will It Continue to Increase?

- Increased prevalence fetal echo
- Adults with ACHD becoming parents - increased risk having affected offspring
- Affected mothers have higher recurrence rate than fathers
 - cytoplasmic inheritance
 - ? more susceptible teratogens

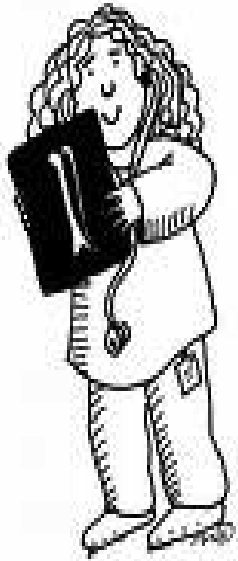


Why Does It Matter?

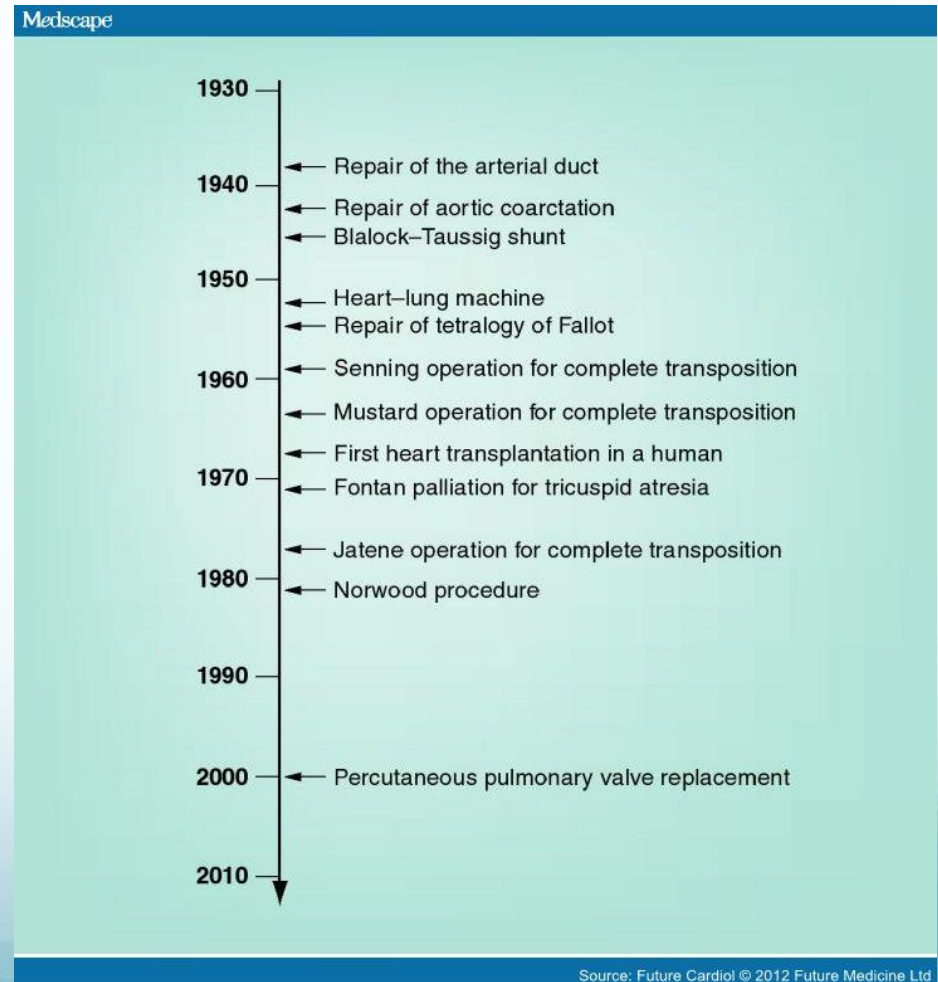
- 85% children born with CHD survive to adulthood
- Now more adults than children with CHD living UK
- Often need long term expert care



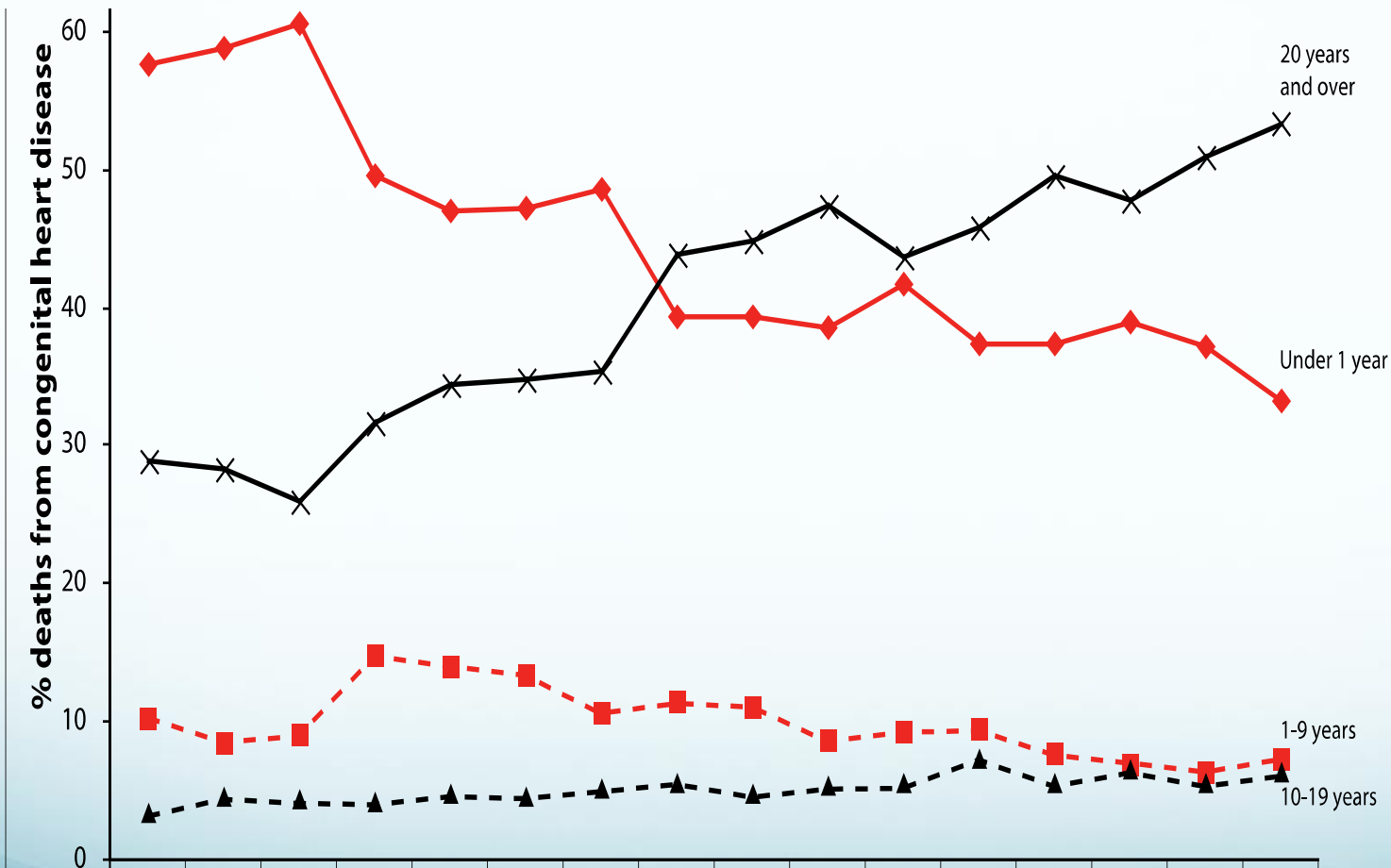
Increased Survival



- Advances imaging
- Medical management
- Surgical advances
- Percutaneous intervention

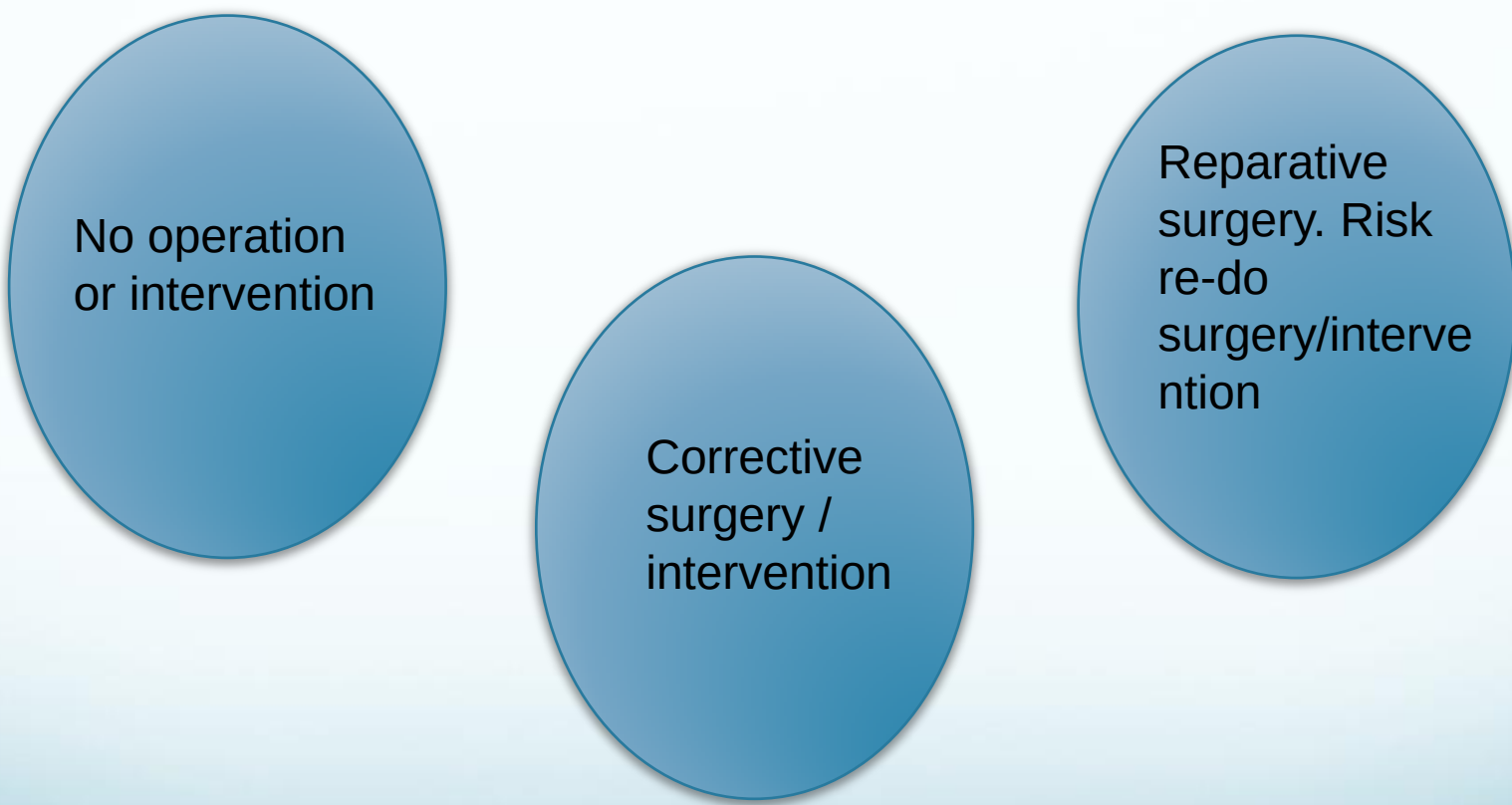


Percentage deaths CHD by Age Group 1986-2001 England & Wales



Death from CHD has shifted from childhood to adulthood

Congenital Heart Disease in Adults



No operation
or intervention

Corrective
surgery /
intervention

Reparative
surgery. Risk
re-do
surgery/intervention

Increased risk: arrhythmia, heart failure, stroke, endocarditis

Congenital Heart Disease in Adults

- Continuous modifications of surgical techniques have changed demographic characteristics adults with CHD
- Challenges natural history
- Many adults remain at risk cardiovascular complications and premature death
- Ageing patients CHD have superimposed acquired disease

Congenital Heart Disease in Adults

- Excess mortality
- Death mainly due cardiovascular causes
 - Heart failure
 - Sudden cardiac death
- 76% deaths in survivors >1year occur in adulthood

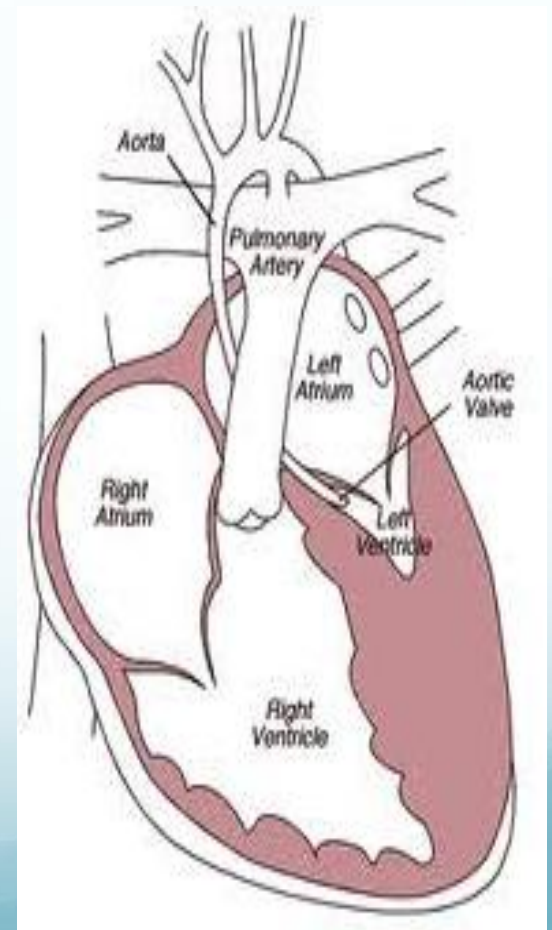
Predictors of Mortality & Influence of Gender

- Age
- Type of defect
 - simple
 - complex
- Number interventions & complications
- 57% CHD occur in women
- Male gender
 - Higher mortality rates in UK & USA in boys & men (5% higher mortality in childhood)

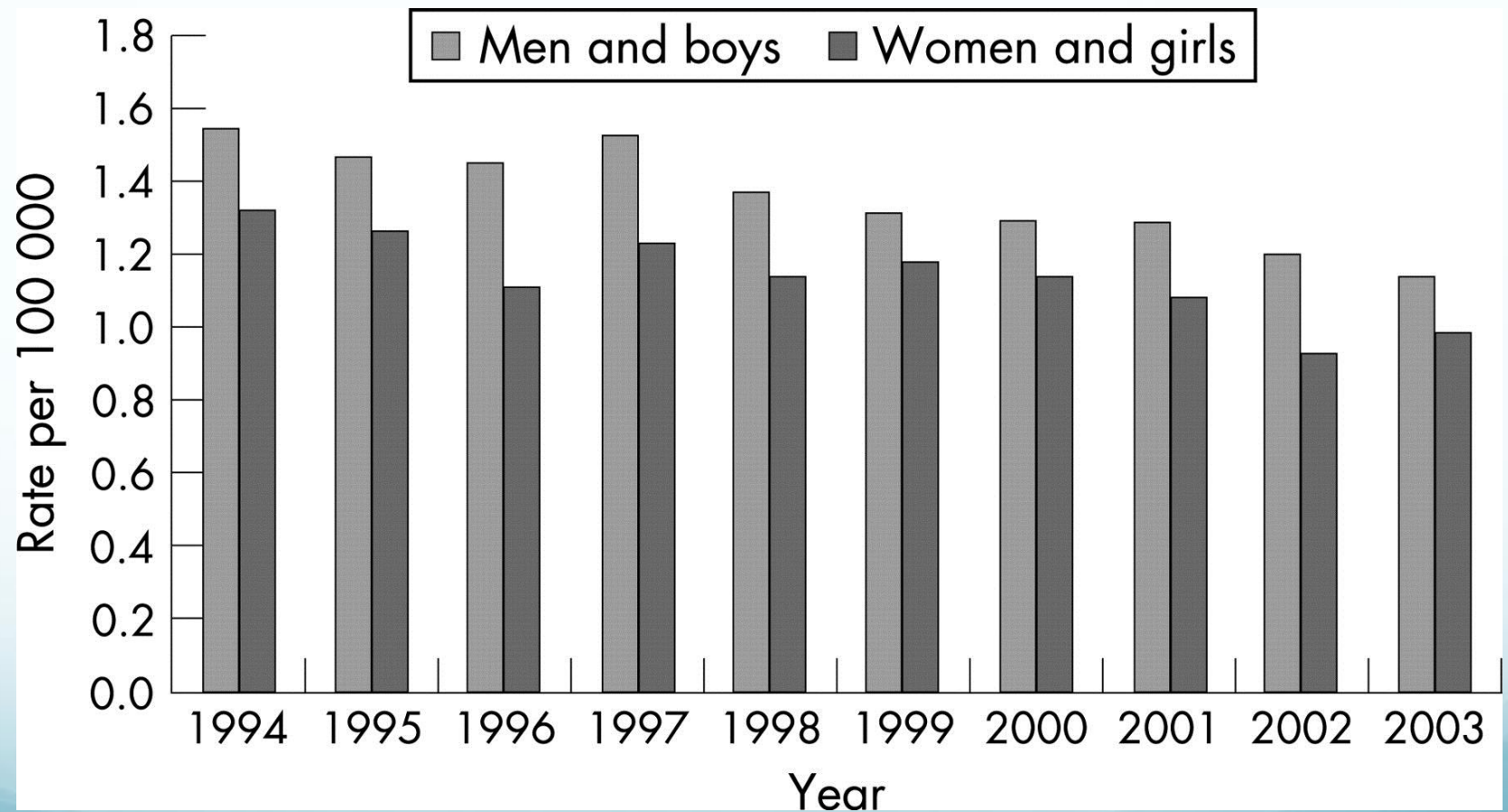
Predictors of Mortality & Influence of Gender

Hypoplastic left heart

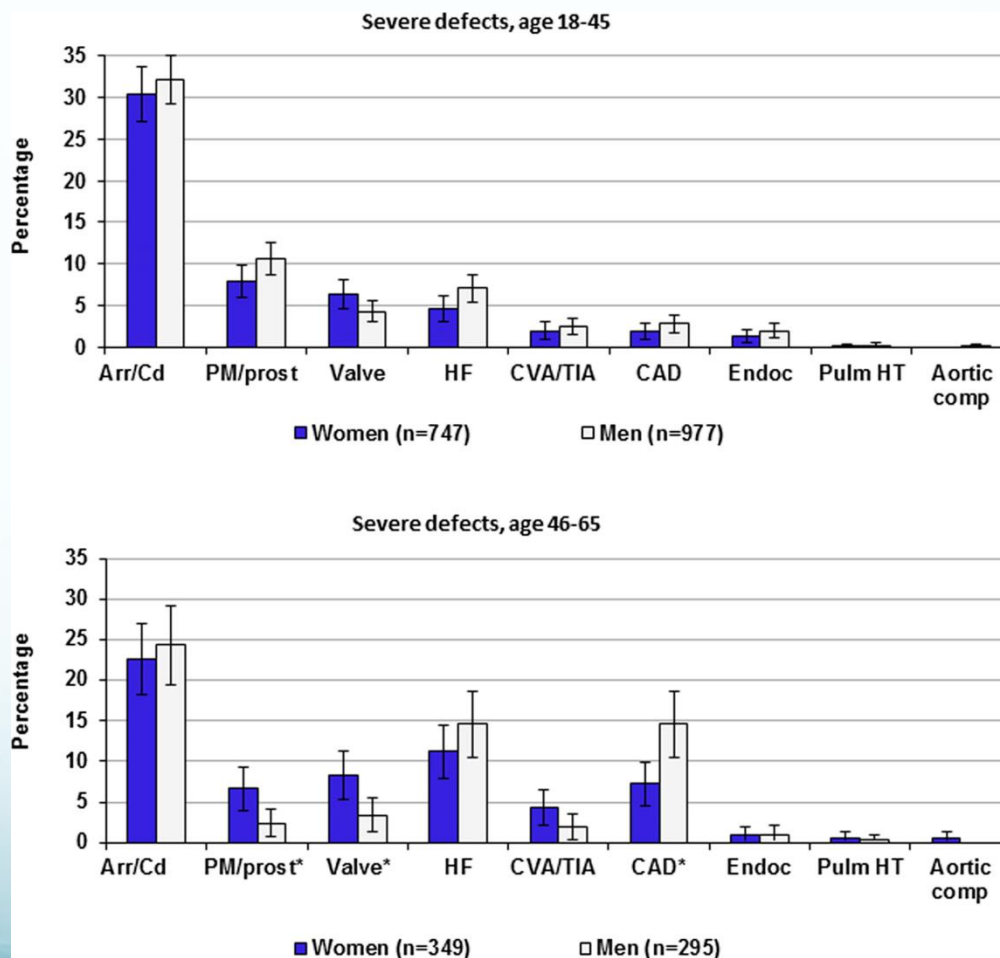
- Female preponderance shunt defects
 - ASD, VSD, PDA, AVCD
- Higher incidence hypoplastic left heart, aortic coarctation, tricuspid atresia, transposition great arteries in men



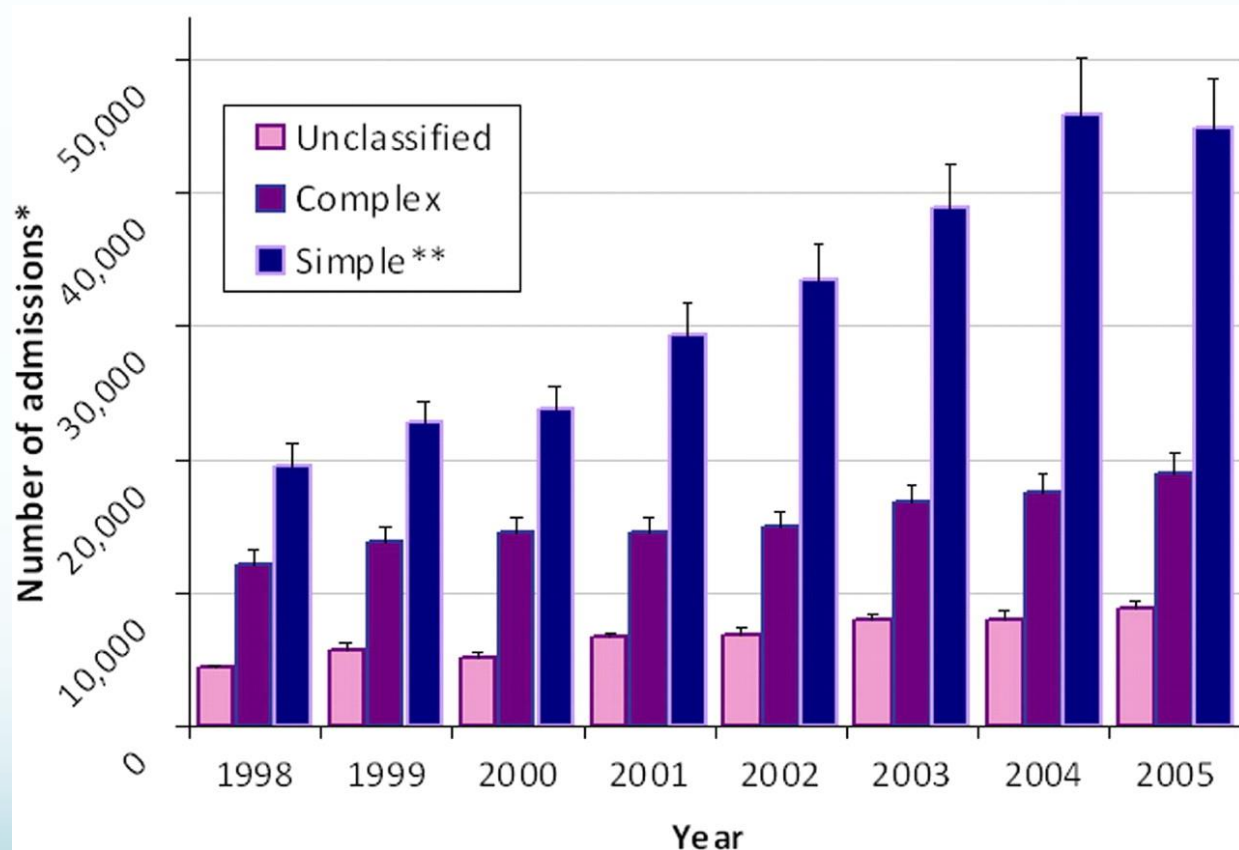
Age-Standardised Death Rates for CHD (underlying cause) England and Wales 1994–2003 by Gender



Gender Differences in Hospital Mortality in Adults With CHD 1996-2005 (Quebec & Netherlands)



Change in Hospital Admissions US



66% increase admissions heart failure

Heart Failure in ACHD



- Although developments in cardiac surgery & interventions have transformed the lives of individuals with congenital heart disease, few are truly curative
- HF is widely prevalent (30% all admissions in ACHD for HF)
- Multifactorial
- Treatment largely empirical
- Randomised trials are long overdue but ACHD functionally & anatomically heterogeneous group

Causes Heart Failure in Patients with CHD

- Previous cardiac surgery – myocardial protection
- Residual Defects
 - Volume overload
 - Pressure overload
- Myocardial Ischaemia
- Systemic right ventricle
- Univentricular circulation (Fontan)
- Arrhythmia
- Permanent pacing
- Acquired heart disease:
 - Coronary disease
 - Hypertension
 - Endocarditis
 - Alcohol / substance misuse

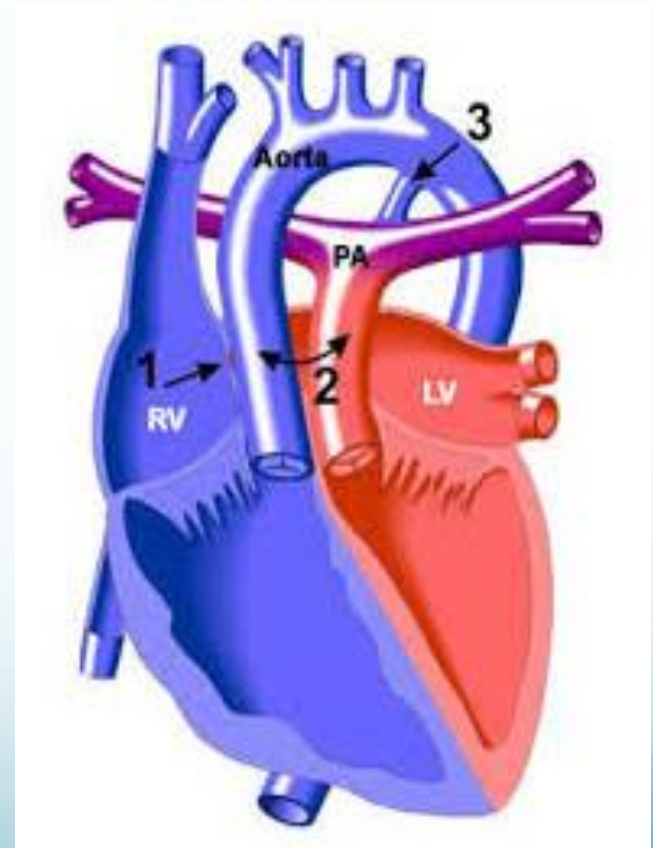


mitral valve and vegetation

- in science generally to solve one set of problems may be to create or discover a whole new set, and of science.....is this more true than of medicine
- George P Elliott 1975 in The American Scholar

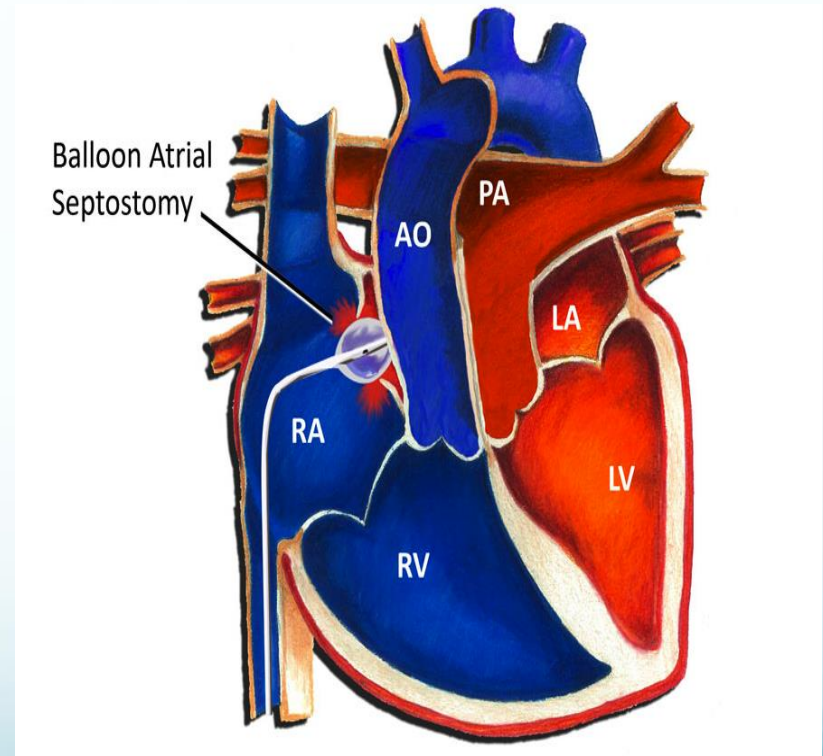
Transposition of the Great Arteries (d-TGA)

- Form cyanotic heart disease
- Great arteries rise from opposite ventricles
- Incompatible with life without intervention



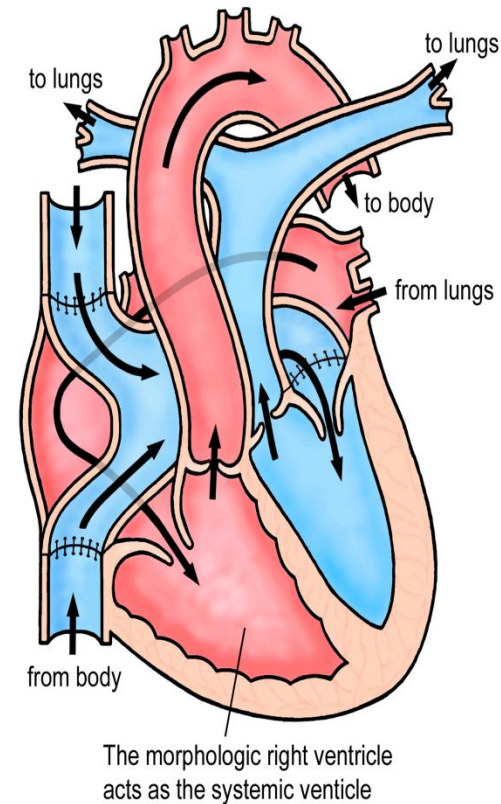
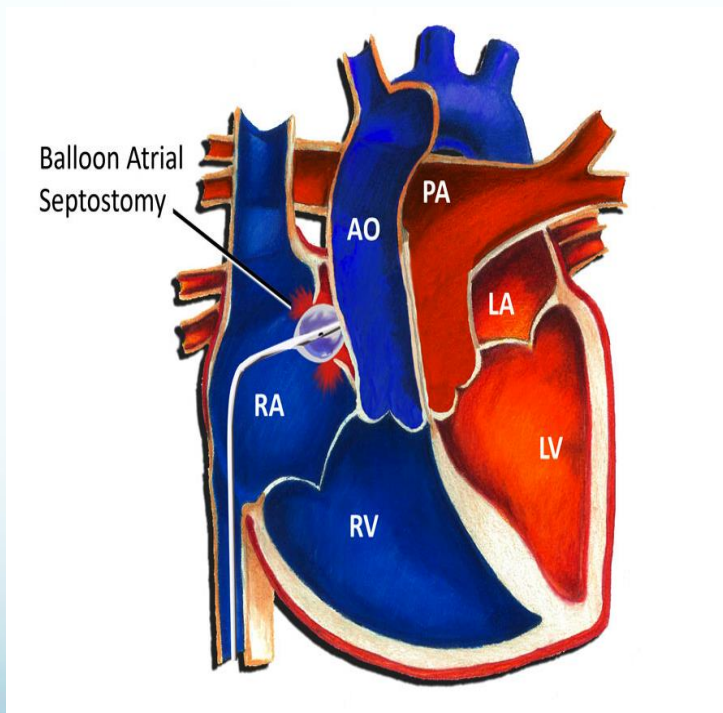
Initial treatment d-TGA

- Babies usually cyanotic in first day of life
- Survive by circulatory mixing either by a patent ductus or by creating a defect in atrial septum by a balloon septostomy (Rashkind balloon)



Surgical Strategies TGA – Atrial Switch

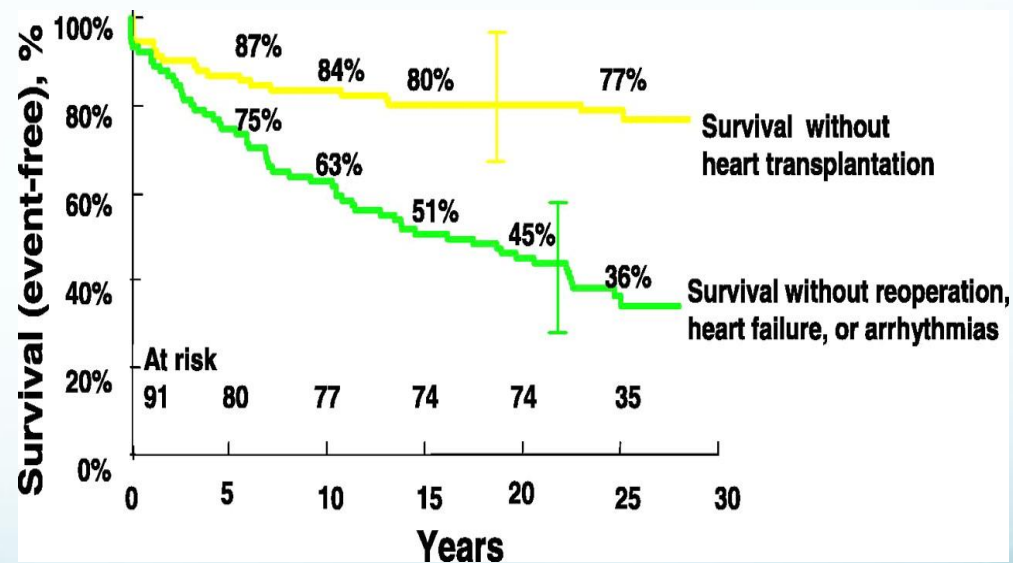
Senning (1958) or Mustard (1964) operation



Late Complications of Atrial Switch

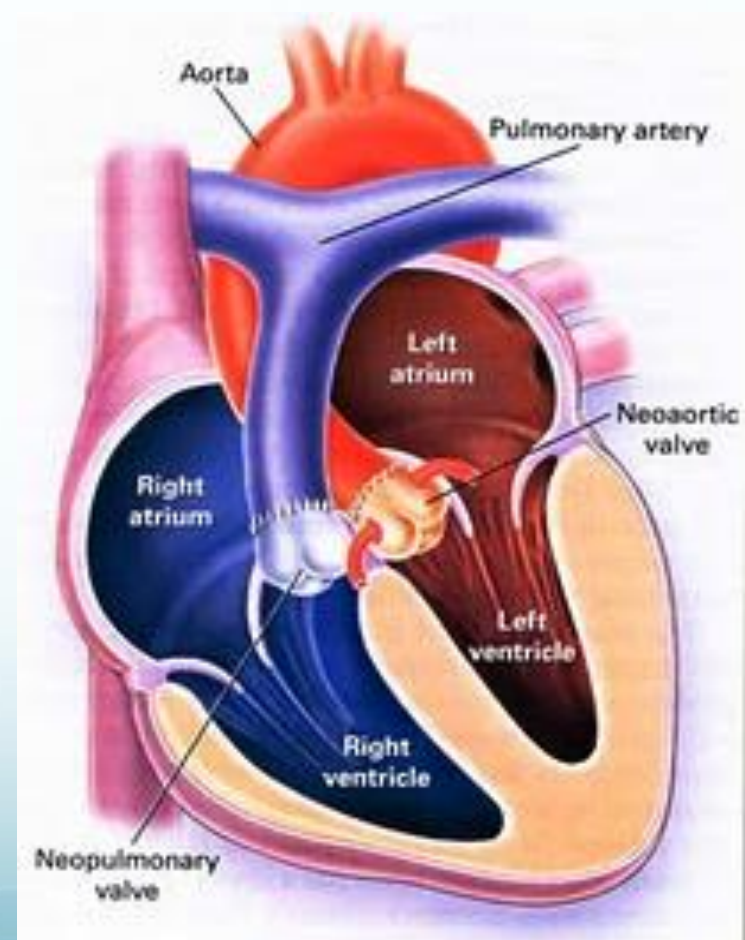
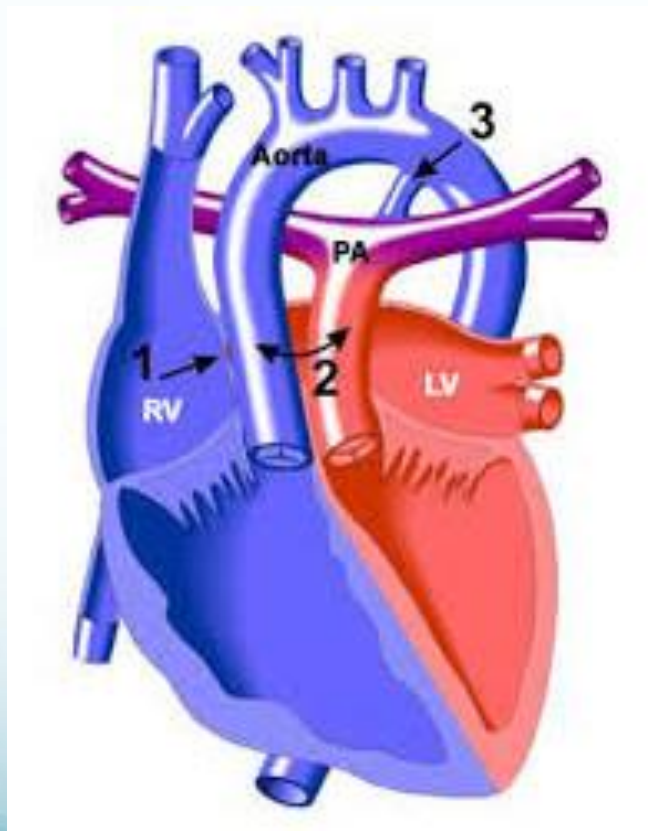
- Arrhythmia
 - sinus node disease (by 20 years 50% patients in sinus rhythm)
 - atrial arrhythmias
- Systemic right ventricle failure
 - morphology of RV
 - ischaemia of hypertrophied RV
- Survival
 - 80% at 14 years
 - 77% at 25 years
 - Death SCD (presumed arrhythmic) & heart failure

Survival (event free) after Mustard operation



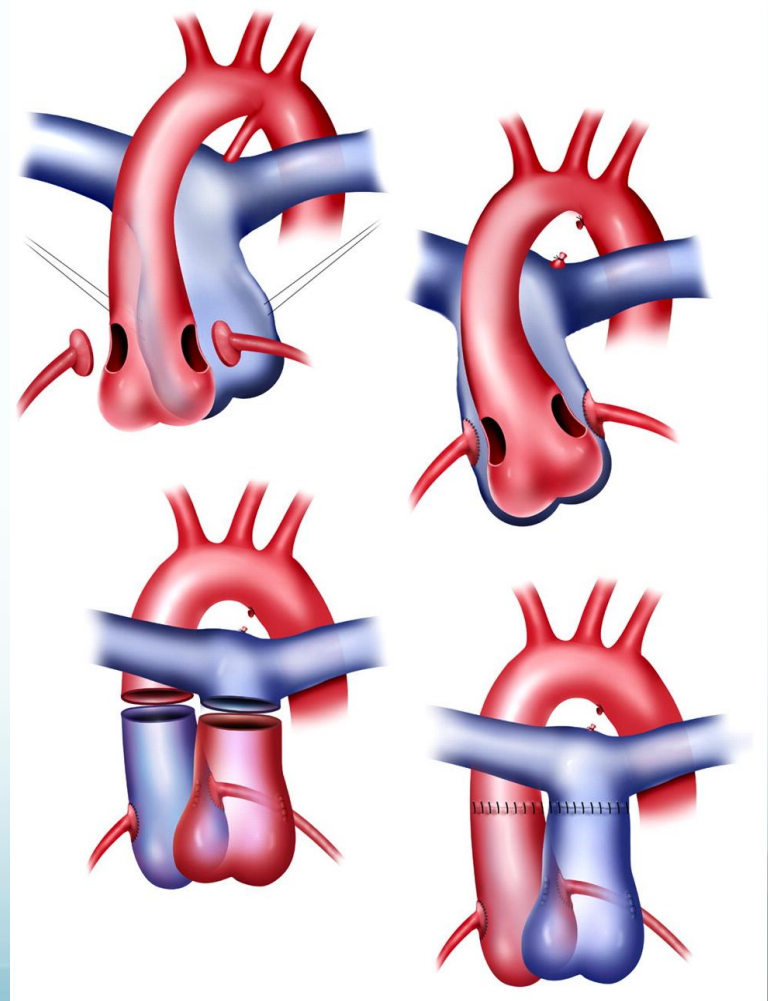
Surgical Strategies TGA – Arterial Switch

Jatene procedure 1976



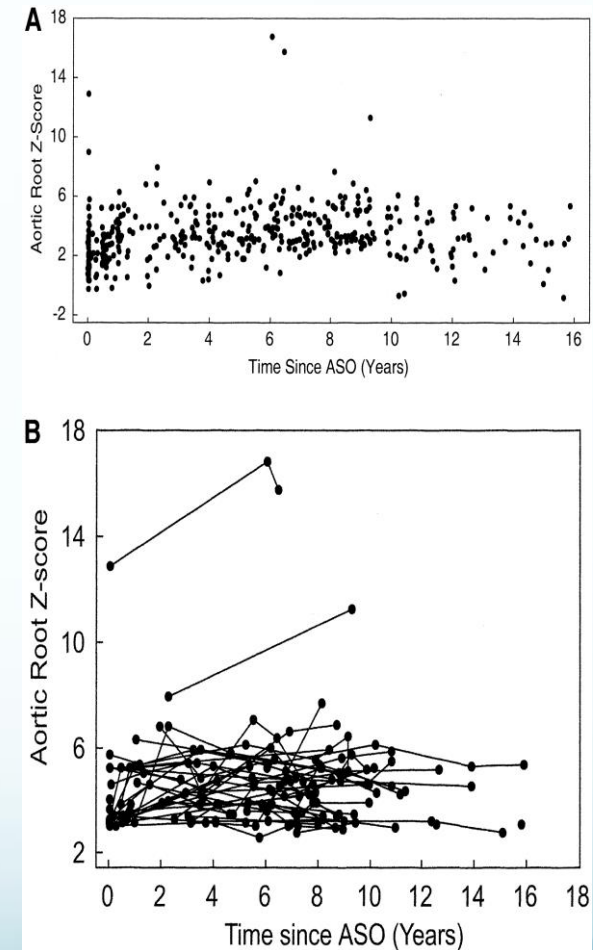
Late Complications of Arterial Switch

- Coronary ischaemia
 - 5 % individuals
- Distortion RV outflow and pulmonary arteries



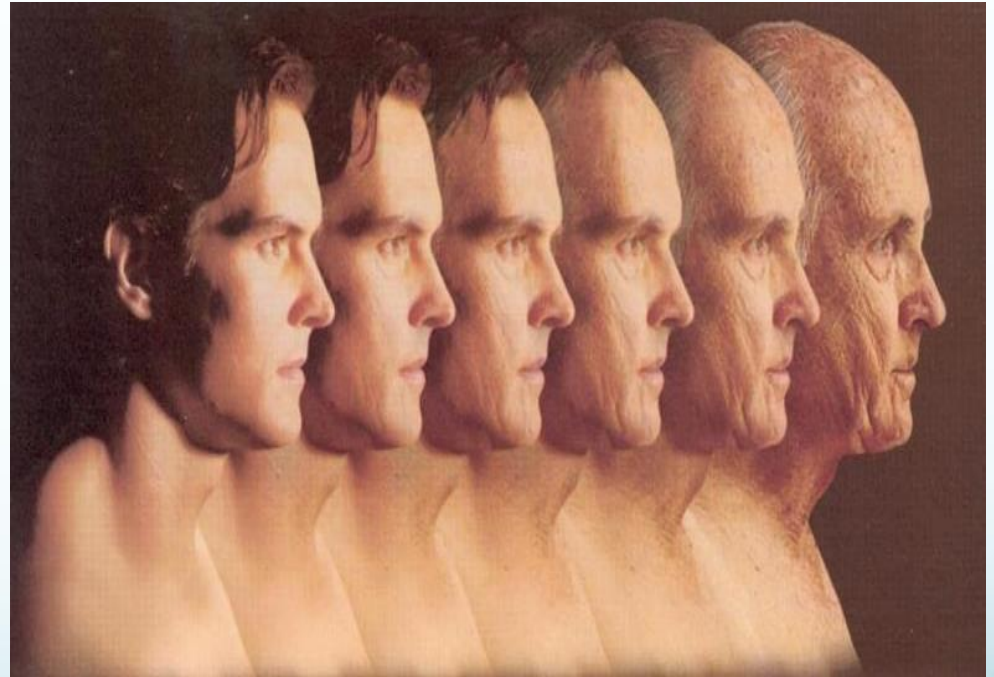
Late Complications of Arterial Switch

- Dilation of neo-aortic root
 - 5 years post op mean aortic Z score 3.6
 - 10 years post op 4.6
- Aortic regurgitation of neo-aortic valve



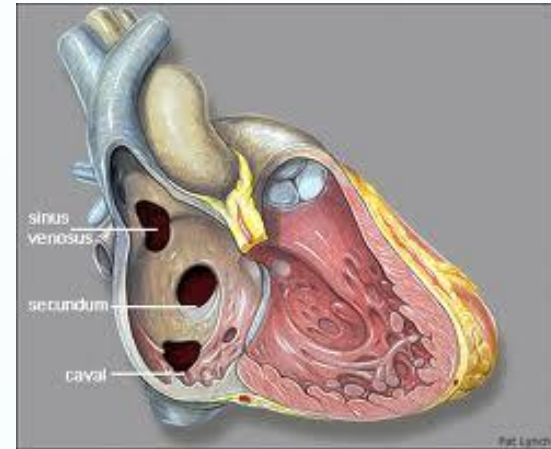
Heart and Ageing

- Myocardial fibrosis
- Coronary atheroma
- Central aortic stiffness
- Reduction endothelial function



Atrial Septal Defect

- 7-10% all congenital defects
- 25% newly diagnosed defects in adulthood
- 3 main types:
 - Secundum (60%)
 - Primum (20%)
 - Sinus venosus (15%)
- Progressive condition



Association secundum & primum ASDs with Trisomy 21
Secundum ASDs skeletal abnormalities forearm & hand (mutations TBX5 – a member of tachyury genes)

Atrial Septal Defect

Presentation

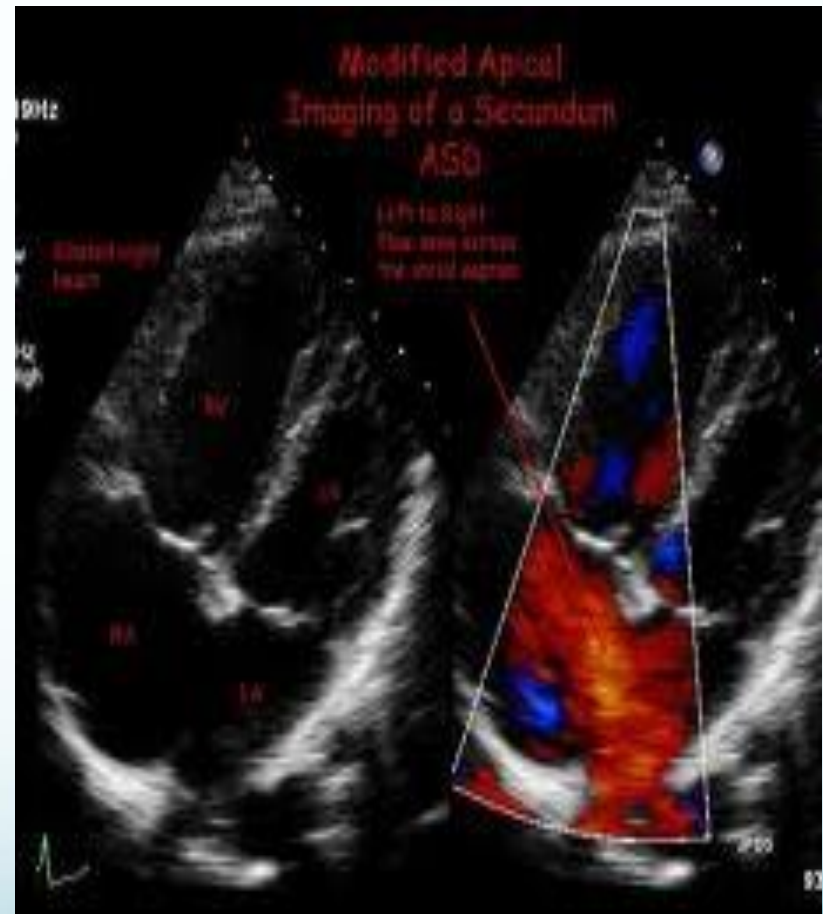
- Children & Adolescents:
 - Frequently asymptomatic in absence PHT (uncommon) or other defects e.g. PS, Ebsteins
- Older Adults:
 - Palpitations
 - SOBE
 - Fatigue
 - Right heart failure
 - Paradoxical embolism

Outlook

- Not benign condition
- Progressive condition
- Causes right heart volume overload
- Increases risk AF
- Closure indicated:
 - haemodynamically significant lesions
 - paradoxical embolism
- Age at closure influences long term outcome

Haemodynamic Effects ASD and Ageing

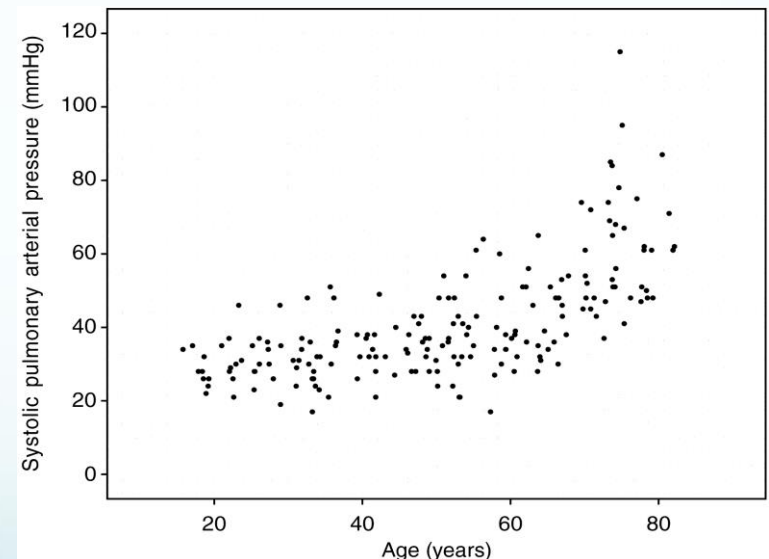
- ASD permits diastolic left to right shunting
- Leads to volume overload
- Right heart dilation
- Extent flow across septum depends on:
 - size defect
 - compliance right and left ventricles



Influence of Age

correlation between PASP and age

- Ageing is intricately linked with symptoms & functional limitation
 - reduction in cardiac output
 - atrial arrhythmias
- Elevation of Pulmonary artery systolic pressure

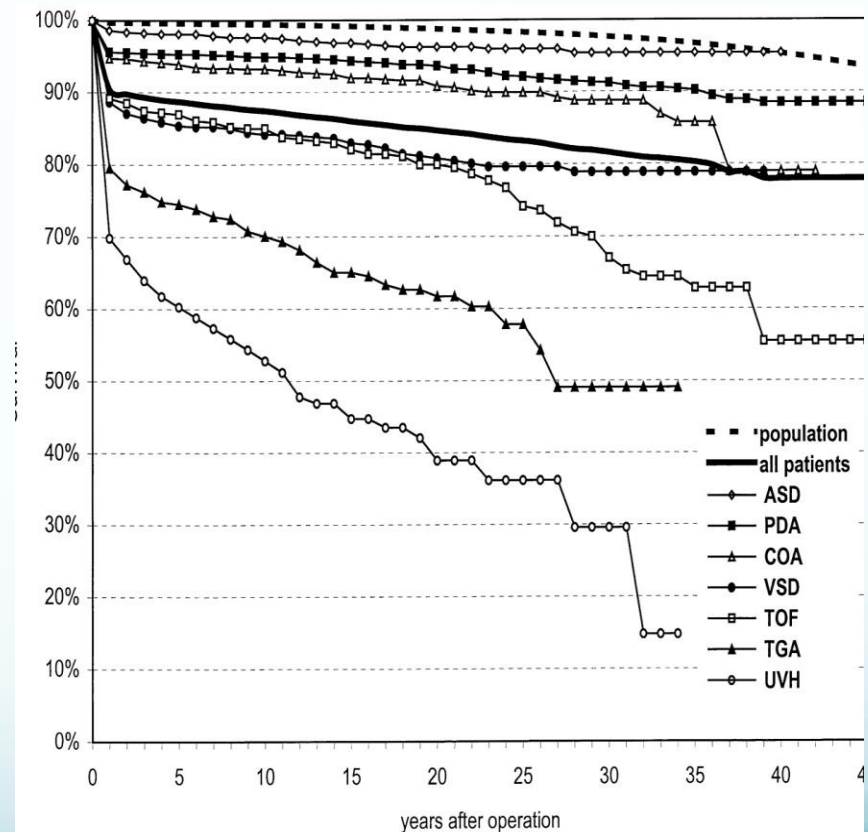


Adults with CHD

- Desire financial security of life insurance
- Difficult group
 - heterogeneous re wide range defects but also treatment and era of treatment
 - increased cardiovascular risk
- Long term mortality studies limited
- Always playing catch up

Finnish Study

- 6461 individuals
- Operated 1953-1989
- Followed up mean 22.3 years (8.8-45.5)
- Late death rate over 45 years 16% compared 7% age, time, gender matched population
- Ranged 5% ASD to 85% univentricular hearts



Broad Brush Strokes for Risk

Positive Prognosticators

- Biventricular circulation
- Systemic left ventricle
- Repair at an early age
- Good functional capacity
- Minor residual haemodynamic lesion

Negative Prognosticators

- Operation in an early surgical era
- Palliation
- Univentricular heart
- Systemic right ventricle
- Arrhythmia history
- PHT
- Multiple or complex residual lesions

In Conclusion

- Adults with CHD are an increasing population as prognosis improves
 - advances medical treatments
 - evolving surgical and percutaneous strategies
- For some conditions long term outlook is poor or uncertain
- Tricky group
- Challenges
 - sparse mortality data
 - not contemporary
 - affects ageing and superimposed acquired heart disease

Any questions?

